Servo Motor

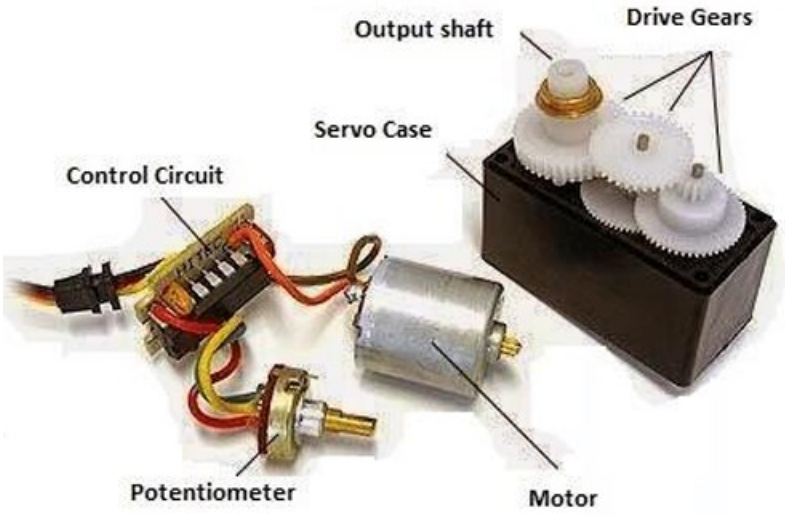


What is Servo?

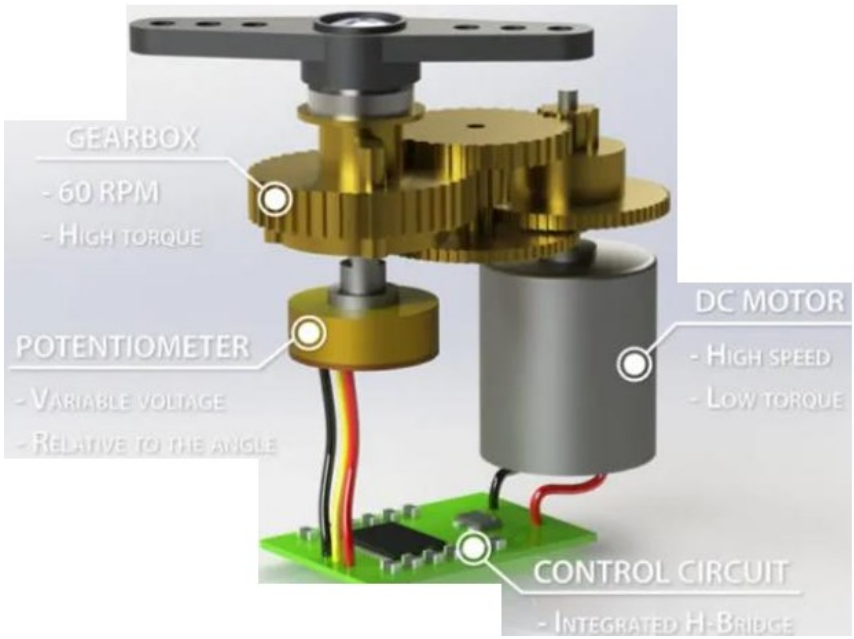
Servo is a general term for a **closed-loop control system**. A closed-loop system uses the feedback signal to adjust the speed and direction of the motor to achieve the desired result.

Servo motor

A **servo motor** is a small device that has an **output shaft**. This shaft can be positioned to specific angular positions by sending the servo a coded signal. As long as the coded signal exists on the input line, the servo will maintain the angular position of the shaft.



There are four main components inside a hobby servo, a **motor**, a **gearbox**, a **potentiometer** and a **control circuit**.



A servo **gear unit** reduces the speed of a motor.

The **potentiometer** provides **position feedback** to the servo control unit where the **current position** of the motor is compared to the **target position**.

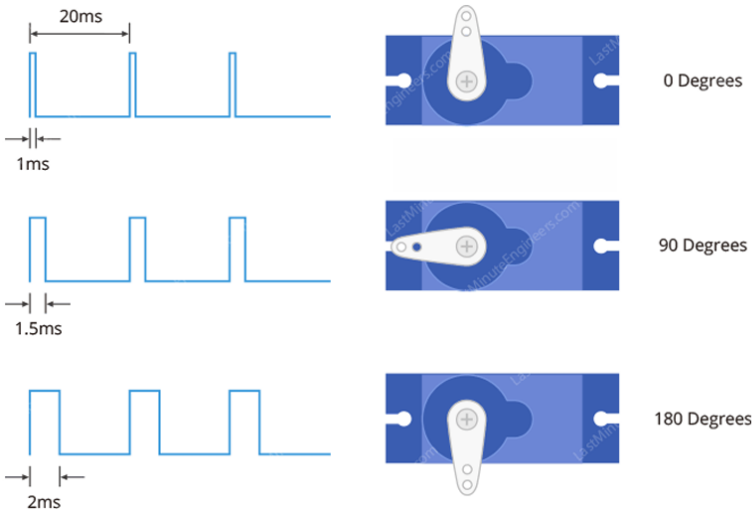
According to the **error**, the **control unit** corrects the actual position of the motor so that it matches the **target position**.

If the **motor** as a controlled device, is **DC motor**, then it is commonly known as a **DC Servo Motor**. If AC operates the controlled motor, it is known as an **AC Servo Motor**.

The main **difference** between the two motors is their **source of power**.

AC servo motors depend on an **AC power source** whereas **DC Servo motors** depend on **DC power source** (like Batteries).

In **AC servo motor** speed is directly proportional to the **frequency of the supply voltage** and the number of magnetic poles, while in **DC servo motor** speed is directly proportional to the **applied voltage**.



- If the pulse is high for 1ms, then the servo angle will be zero.
- If the pulse is high for 1.5ms, then the servo will be at its center position.
- If the pulse is high for 2ms, then the servo will be at 180 degrees.
- Pulses ranging between 1ms and 2ms will move the servo shaft through the full 180 degrees of its travel.

Note:
The minimum and maximum duration of the pulses can sometimes vary with different brands and they can be 0.5ms for 0 degrees and 2.5ms for 180 degrees position.

Servo Motor Pinout

Servo motors typically have three connections and are as follows:



GND is a common ground for both the motor and logic.
5V is a positive voltage that powers the servo.
Control is input for the control system.

The color of the wires varies between servo motors, but the red wire is always 5V and GND will either be black or brown. The control wire is usually orange or yellow.